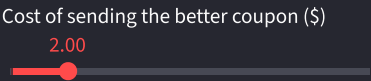
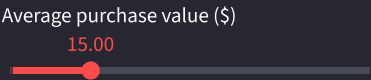


Business assumptions



- Targeting rule
- ☒ Value threshold
 - ☐ Budget cap

Filter held-out customers

Coupon type

Bar x

Carry out & Take a... x

Coffee House x

Restaurant(20-50) x

Restaurant(<20) x

Destination

Home x

No Urgent Place x

Work x

Income bracket

\$50000 - \$62499 x

\$62500 - \$74999 x

\$75000 - \$87499 x

\$87500 - \$99999 x

Causal ML for Coupon Targeting

Which customers should receive a 1-day coupon instead of a 2-hour coupon? Every reported causal estimate and targeting result below is computed on a held-out evaluation set, not the rows used to train the models.

Overview & Diagnostics Top Customers Customer Lookup

Showing 3,153 filtered customers from a 3,153-row held-out evaluation set.

Customers to target	Incremental revenue	Campaign cost	Campaign ROI
1,790	\$5,382	\$3,580	50.3%

Held-out causal estimates

T-learner average uplift	Propensity matching ATT	Doubly-robust AIPW ATE
+0.149	+0.159	+0.168
↑ 95% CI [+0.129, +0.159]	↑ 95% CI [+0.102, +0.233]	↑ 95% CI [+0.131, +0.199]

The estimators use different assumptions. Agreement is reassuring, but it does not by itself prove that all confounding has been removed.

Positivity, overlap, and treatment-balance checks

Control share (2h)

Treated share (1d)

Propensities clipped

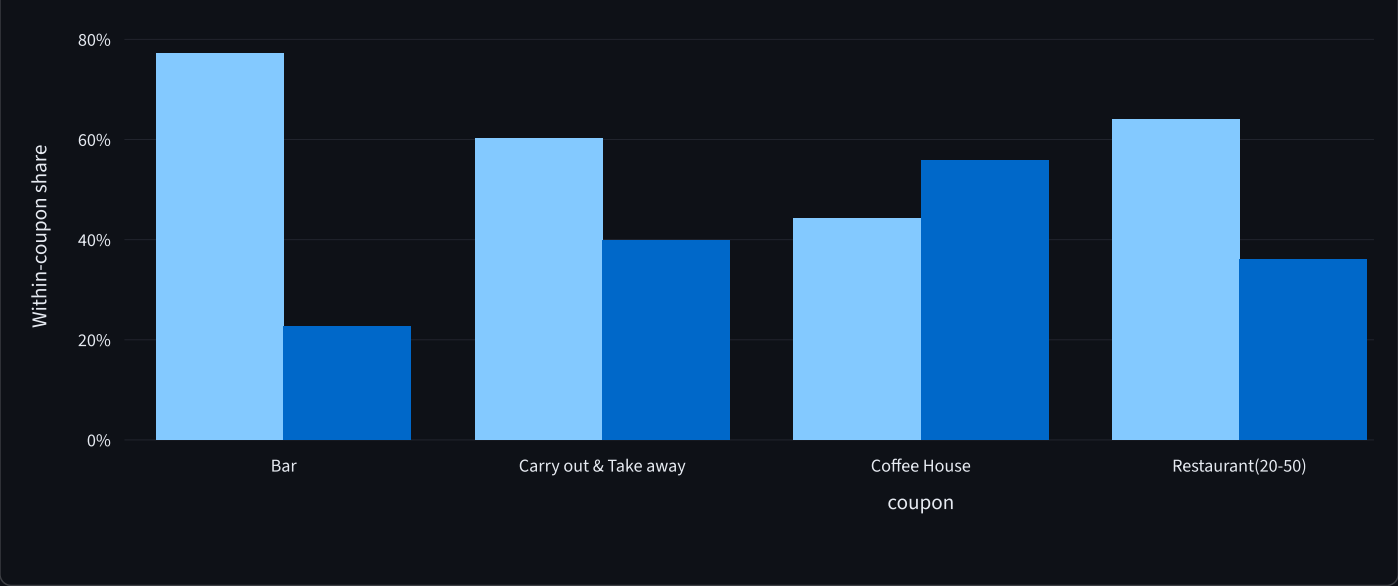
44.2%

55.8%

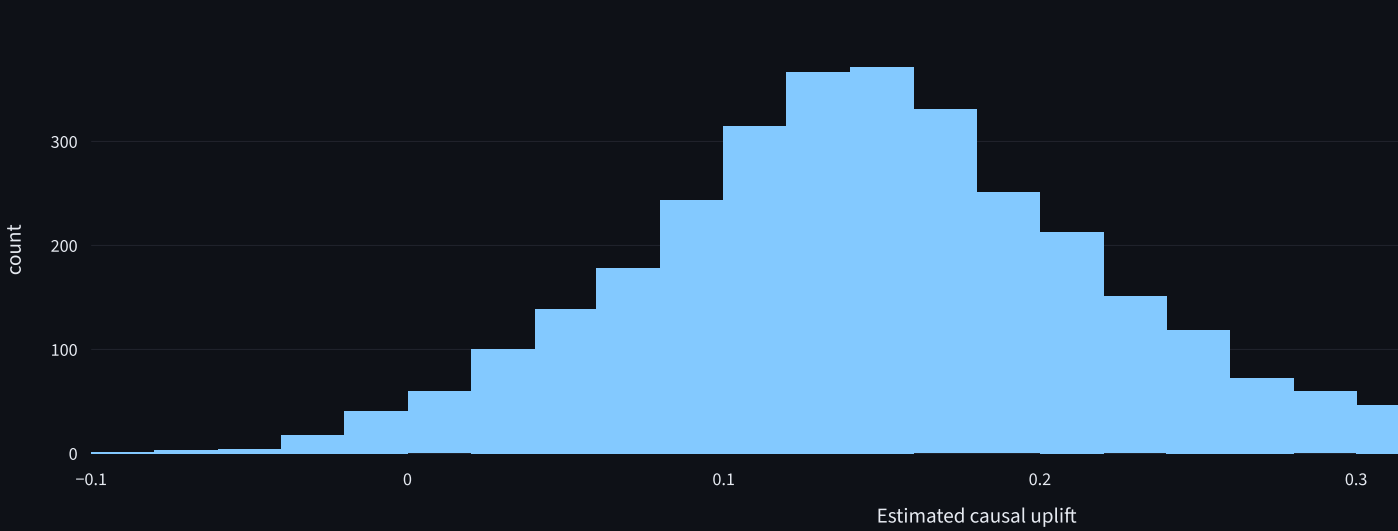
0 (0.0%)

No coupon type exceeds 90% concentration in a single expiration arm.

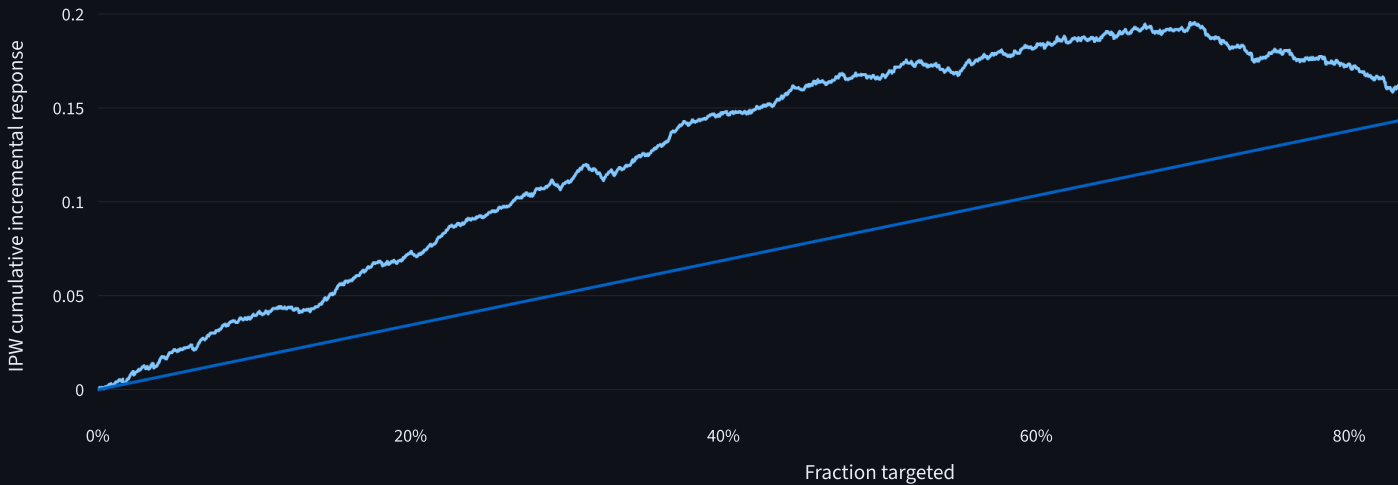
Expiration mix within each coupon type



Held-out distribution of estimated individual uplift



Held-out uplift/Qini-style cumulative gain



Held-out average uplift by coupon type

